

# Beyond Persistence? PAGE: Persistence-Anchored Gated Forecasting with Spatio-Temporal Graph Networks for Dengue

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**Abstract**—District-level dengue forecasts three weeks ahead give Sri Lankan health services time to act. Spatio-temporal graph neural networks (ST-GNNs) and SEIR-informed models have been proposed for this task, but they predict the case level directly and are evaluated without the persistence forecast, even though last week’s count explains  $r^2 = 0.89$  of the next. We propose PAGE (Persistence-Anchored Gated Estimation), a forecasting head for any spatio-temporal encoder that predicts a gated departure from the last observed week, at the cost of one parameter. We correct the benchmark into a leakage-free 559-week dataset, extend it to 2026 by parsing 127 new reports, and evaluate five published ST-GNNs and an LSTM under one protocol. PAGE repairs the three encoders that fail as published by 5.95–16.29 validation RMSE (9/9 paired runs each), and with AAGCN it beats persistence on all nine runs on validation and test. An ablation attributes the gain to the anchor and the gate; an SEIR simulator in the same head adds nothing, and on 85 later forecast starts it ties its no-SEIR twin ( $p = 0.67$ ). On those 2024–2026 weeks no learned model beats persistence on RMSE (34.75 against 35.43 for PAGE), although PAGE has lower MAE and SMAPE. For this target, persistence is the prior that matters. Code and data: <https://anonymous.4open.science/r/TOD0>.

**Index Terms**—dengue forecasting, spatio-temporal graph neural networks, persistence baseline, physics-informed learning, data leakage, prospective evaluation

## I. INTRODUCTION

Dengue remains a major public-health problem in Sri Lanka, with recurrent outbreaks that place pressure on hospitals and vector-control services; the 2017 epidemic alone recorded more than 186,000 cases and 440 deaths [1]. Reliable district-level forecasts can support earlier preparation of clinical resources and targeted control. We consider weekly forecasting for the 25 districts of Sri Lanka with a three-week horizon.

Recent work follows two main directions. Spatio-temporal graph neural networks (ST-GNNs) represent districts as nodes and combine spatial relationships with temporal case patterns; Weng et al. [2] evaluated several such architectures on Sri Lankan dengue data. A second direction introduces epidemic mechanisms: Liu et al. [3] use an LSTM to predict the force of infection of an SEIR model.

However, an important property of weekly dengue incidence is easily overlooked: the recent case count is a very strong predictor. Cases in week  $t - 1$  explain  $r^2 = 0.89$  of the variation in week  $t$ , whereas the best climate covariate explains only  $r^2 = 0.03$ , so the persistence forecast  $\hat{y}_{t+h} = y_t$  is a strong baseline. Existing ST-GNNs nevertheless predict the future case level directly, so they must first learn the identity

$\hat{y}_{t+1} \approx y_t$  before learning any useful correction. Three of five published ST-GNNs fail to do this and are 5–17 RMSE worse than persistence on validation. The existing benchmark hides this weakness, because it reports no persistence baseline and its data leak future information (Section III). Mechanistic approaches also do not isolate the contribution of the physics with an otherwise identical model.

To address this problem, we propose PAGE (Persistence-Anchored Gated Estimation), a lightweight head that can be attached to any spatio-temporal encoder. PAGE begins with the persistence forecast and learns only a departure from it, scaled by a learned gate initialised close to zero, so the model starts from a strong forecast and moves away from persistence only when the data support it. The same head lets an SEIR simulator generate the departure, which gives a matched test of whether epidemic physics adds value.

PAGE reduces the validation RMSE of STGAT, A3TGCN and DCRNN by 5.95–16.29 (9/9 paired runs each), and with AAGCN it beats persistence on all nine runs on validation and test. Ablations show that the gain comes from the anchor and the gate, not the SEIR simulator. On later 2024–2026 weeks the SEIR and non-SEIR variants tie ( $p = 0.67$ ), and persistence has the lowest RMSE (34.75 against 35.43), although PAGE has lower MAE and SMAPE.

Our contributions are:

- 1) PAGE, an encoder-independent persistence-anchored gated head with one extra parameter, with ablations that separate the anchor, the gate and the SEIR component.
- 2) An audit of the Sri Lankan dengue benchmark and a corrected, leakage-controlled 559-week dataset (2013–2024), extended with 127 newly parsed reports to August 2026.
- 3) A common evaluation of six neural encoders with matched physics and non-physics controls, paired tests and a prospective 2024–2026 test, showing that persistence is a stronger prior for this problem than the tested SEIR dynamics.

## II. RELATED WORK

**ST-GNNs for epidemics.** Spatio-temporal graph networks couple a graph operator over regions with a temporal operator over history. STGAT and A3TGCN [4] use attention, AST-GCN [5] attends over space and time, AAGCN [6] adapts its adjacency, DCRNN [7] diffuses over the graph inside a

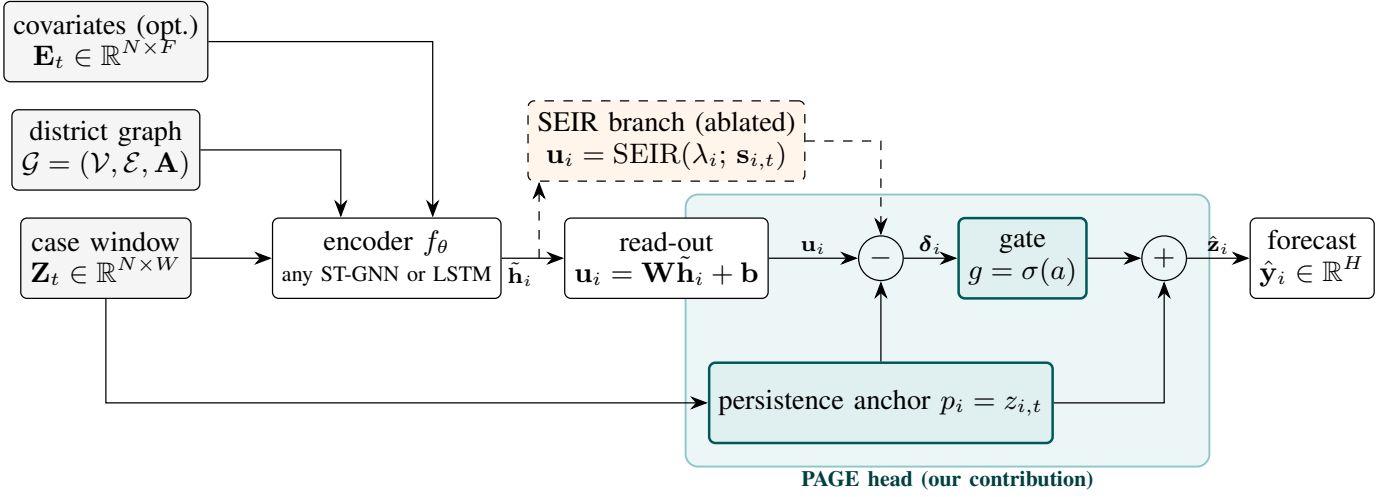


Fig. 1. PAGE. An unchanged encoder reads the last  $W$  weeks of standardised log cases on the district graph. Its read-out  $\mathbf{u}_i$  is turned into a departure  $\delta_i = \mathbf{u}_i - p_i \mathbf{1}$  from the persistence anchor, shrunk by a learned gate, and added back to the anchor. The dashed SEIR branch replaces the read-out with a differentiable simulator; it is the component the ablation removes.

recurrent cell, and Graph WaveNet [8] learns the adjacency from node embeddings. Cola-GNN [9] and EpiGNN [10] adapt such models to epidemic forecasting with cross-location attention and transmission-risk encodings. Weng et al. [2] benchmark five traffic ST-GNNs on Sri Lankan dengue, and GulMohamed et al. [11] add per-horizon heads and temporal attention. These models predict the case level directly, and the dengue benchmark reports no persistence baseline.

**Mechanistic hybrids.** SEIR-LSTM [3] predicts the force of infection of an SEIR model with an LSTM. EINN [12] share latent dynamics between a neural and a mechanistic model, while CausalGNN [13] and MepoGNN [14] embed a metapopulation SEIR in a graph network. Physics-informed networks [15], [16] add the dynamics as a loss. These works compare against purely data-driven models, but not against the same head with the simulator removed, so the share of the gain due to physics is not measured.

**Simple baselines.** In classical forecasting the naive forecast is the reference every method must beat, and modelling the change rather than the level is the standard treatment of a near-random-walk series [17]. In deep spatio-temporal forecasting, STID [18] shows that identity embeddings can match graph operators. PAGE brings the classical idea into the ST-GNN head.

### III. DATASET CONSTRUCTION

**Auditing the benchmark.** The array used by the existing benchmark [2] has no date axis. Dating each row by matching its 25 district counts against the published Weekly Epidemiological Reports (WER) shows that rows 0–47 are from 2023 and rows 51 onward run 2013–2022, so 45 rows from 2023 sit in the training split of every fold that tests on 2018–2022. The weather channels were not reordered with the cases: the array’s temperature matches ERA5 at  $r = 0.95$  on its own row order but  $r = 0.45$  on the dates the case rows carry, and its “lag-12” rainfall matches ERA5 rain twelve weeks *later* ( $r = 0.68$ ,

against 0.19 twelve weeks earlier). Its largest spike, a 19-fold jump in one week, is a dragged spreadsheet formula in one printed table (15 of 24 cells equal the sum of the two before). Two of these findings leak future information, so scores on this array are optimistic.

**The corrected dataset.** The 2013–2024 case counts come from the benchmark authors’ table of parsed WER rows; we do not re-parse those reports. We re-date every week by its report volume and number, map the reporting divisions to the 25 districts, and correct the spreadsheet error from the published PDF. We add ERA5 daily weather and census population ourselves, and record a SHA-256 checksum for every input file. The result covers  $N = 25$  districts over 559 weeks (2013-W26 to 2024-W10), with weekly cases, temperature, precipitation, humidity, soil moisture and population. Seven weeks with no report stay missing and are never interpolated. A forecast for week  $t$  reads cases up to  $t - 1$  and climate up to  $t - 2$  (ERA5 is released about five days late), and a perturbation test asserts that no later value can change a forecast. The district graph is the benchmark’s adjacency list with self-loops (141 directed entries); the prospective test uses a corrected version with two non-border entries removed (139).

**Extension to 2026.** For a prospective test we parse the 127 WERs from 2024-W11 to 2026-W32 under quality rules committed before any count was read: district totals must equal the national total, and a dragged-formula pattern or a reprinted table voids a week, which is replaced by the difference of the cumulative rows or left missing. This gives 109 observed, 8 corrected and 10 missing weeks. A first scoring run exposed two reconstruction defects (the first report of a year carries the previous year’s total, and two reports reprint the previous week’s table). The rules were amended on report semantics alone, with no model setting changed, so we call this an *amended* prospective evaluation.

#### IV. THE PAGE FRAMEWORK

##### A. Problem Formulation

Let  $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{A})$  be the district graph with  $|\mathcal{V}| = N$  and adjacency  $\mathbf{A} \in \{0, 1\}^{N \times N}$ , and let  $y_{i,t} \geq 0$  be the cases reported by district  $i$  in week  $t$ . We model the standardised log count  $z_{i,t} = (\log(1 + y_{i,t}) - m)/s$ , with  $m$  and  $s$  computed on training weeks only. At forecast origin  $t$  the input is the window  $\mathbf{Z}_t = [z_{i,t-W+1}, \dots, z_{i,t}]_{i \in \mathcal{V}} \in \mathbb{R}^{N \times W}$  and, optionally, covariates  $\mathbf{E}_t \in \mathbb{R}^{N \times F}$  at causal lags. The task is to predict  $y_{i,t+h}$  for every district and horizon  $h = 1, \dots, H$ .

##### B. Encoder

Any spatio-temporal encoder  $f_\theta$  maps the window to one representation per district,  $\mathbf{H} = f_\theta(\mathbf{Z}_t, \mathbf{A}) \in \mathbb{R}^{N \times d}$ , and covariates are concatenated around it,  $\mathbf{h}_i = [\mathbf{h}_i \| \mathbf{e}_{i,t}]$ , so that a published architecture runs exactly as designed. A linear read-out gives  $\mathbf{u}_i = \mathbf{W}\mathbf{h}_i + \mathbf{b} \in \mathbb{R}^H$ . Published models output  $\hat{\mathbf{z}}_i = \mathbf{u}_i$ ; we call this the *direct* head.

##### C. Persistence-Anchored Gated Head

PAGE anchors every horizon on the last observed week,  $p_i = z_{i,t}$ , and lets the network supply only a gated departure from it (Fig. 1):

$$\delta_i = \mathbf{u}_i - p_i \mathbf{1}_H, \quad \hat{\mathbf{z}}_i = p_i \mathbf{1}_H + g \delta_i, \quad g = \sigma(a), \quad (1)$$

where  $a$  is one learned scalar initialised at  $a_0 < 0$ , so training starts near the persistence forecast. Equivalently,  $\hat{\mathbf{z}}_i = (1 - g)p_i \mathbf{1}_H + g \mathbf{u}_i$  is a learned convex combination of persistence and the encoder's forecast. Counts are recovered as  $\hat{y}_{i,t+h} = \max(0, \exp(s\hat{z}_{i,h} + m) - 1)$ .

Three properties follow. (i) *The identity is free*: at  $g \rightarrow 0$  the model is exactly persistence, which a direct head must instead carry through its message passing and recurrence. (ii) *The gate is a learned shrinkage*:  $g$  grows only as far as  $\mathbf{u}_i$  improves on  $p_i$ , so a weak encoder cannot pull the forecast far from a strong baseline. (iii) *It costs one parameter* and no extra computation. Setting  $g \equiv 1$  with an unconstrained departure gives the *residual* head, and removing the anchor gives the *direct* head; both are ablations.

##### D. Optional SEIR Branch

To test whether transmission dynamics help, the read-out can be replaced by a differentiable SEIR model. The encoder predicts a force of infection  $\lambda_{i,h} = \min(\exp(\mathbf{w}^\top \tilde{\mathbf{h}}_i + b_\lambda), \lambda_{\max})$ , and  $\dot{S} = -\lambda S$ ,  $\dot{E} = \lambda S - \omega E$ ,  $\dot{I} = \omega E - \gamma I$ ,  $\dot{R} = \gamma I$  are stepped daily from a state  $\mathbf{s}_{i,t}$  seeded by the observed cases of weeks  $t-1$  and  $t$  divided by a reporting rate  $\rho$ . The weekly flow  $E \rightarrow I$ , scaled by  $\rho$  and the district population, gives counts whose standardised log replaces  $\mathbf{u}_i$  in Eq. (1). With the anchor and gate this is the SEIR-GNN (SEIR-LSTM when  $f_\theta$  is an LSTM); without them it is a pure SEIR decoder.

##### E. Learning Objective

All parameters  $\Theta = \{\theta, \mathbf{W}, \mathbf{b}, a\}$ , plus  $\mathbf{w}$ ,  $b_\lambda$  and learned scales on the seeded state and  $\rho$  when the SEIR branch is used, minimise the squared error on the standardised scale over a mini-batch  $\mathcal{B}$  of origins,

$$\mathcal{L}(\Theta) = \frac{1}{|\mathcal{B}|NH} \sum_{t \in \mathcal{B}} \sum_{i=1}^N \sum_{h=1}^H (\hat{z}_{i,h}^{(t)} - z_{i,t+h})^2, \quad (2)$$

with early stopping on validation RMSE in counts. We also test a negative-binomial (NB) likelihood, because  $\exp$  of a squared-error fit on  $\log(1 + y)$  estimates a conditional median while RMSE rewards the conditional mean.

##### F. Implementation Details

$N = 25$ ,  $W = 3$ ,  $H = 3$ ,  $d = 64$ ,  $a_0 = -2$  ( $g_0 = 0.12$ ). SEIR rates are per day:  $\omega = 0.1$ ,  $\gamma = 1/7$ ,  $\rho = 1/11$ ,  $\lambda_{\max} = 1/7$ , with stage durations from [19] and an initial susceptible fraction of 0.318 from a serosurvey. Training uses Adam (learning rate  $3 \times 10^{-3}$ , weight decay  $10^{-4}$ ), batch size 32, dropout 0.1, at most 300 epochs (400 for NB), early stopping with patience 40, and the best-validation weights restored; seeds 0, 1, 2. All models are written in plain PyTorch with dense  $N \times N$  graph operations, and each published encoder is checked weight-for-weight against its reference implementation. The main study (810 runs) ran in 3.0 hours on a 4-core Kaggle CPU; the prospective study ran on a laptop CPU (Intel Core i5-12450H). No GPU is needed.

#### V. EXPERIMENTAL SETUP

**Dataset.** All development results use the corrected 559-week series (Section III); the 2024–2026 weeks are used only for the prospective test. Every model, baseline and control is scored on exactly the same validation and test windows and trained on the same training windows, except that arms with climate lags skip the first training weeks, which have no climate history that far back. Windows that touch a missing week are dropped for all of them.

**Baselines.** Persistence  $\hat{y}_{t+h} = y_t$  and seasonal naive  $\hat{y}_{t+h} = y_{t+h-52}$ ; the mandatory sequence and graph baselines, the two-layer LSTM of SEIR-LSTM [3] and a GCN [20] against the same network without message passing; the five ST-GNNs of the Sri Lankan benchmark [2] (STGAT, A3TGCN, ASTGCN, AAGCN, DCRNN) with their published direct heads; SEIR-informed heads (pure SEIR decoder, SEIR-LSTM, SEIR-GNN); and non-neural learners: gradient-boosted trees [21] with and without climate, a negative-binomial GLM,  $k$ -nearest-neighbour analogues, an AR(3) ridge regression and an STID-style MLP [18].

**Protocol.** Rolling origins at 0.55, 0.70 and 0.85 of the series, with three seeds each; the 30 windows before each origin are validation and the next 15% are test, with no shuffling across time. Selection uses validation RMSE only, and test numbers never choose a configuration. The protocol, and for each experiment its arms, control and decision rule, were committed before it ran. Differences are paired on matched (origin,

TABLE I

COMPARATIVE ANALYSIS ON ONE HARNESS (CORRECTED DATA, 3 ORIGINS  $\times$  3 SEEDS). MEAN VALIDATION / TEST RMSE IN WEEKLY CASES. TOP: THE FIVE PUBLISHED ST-GNNs AND THE LSTM OF SEIR-LSTM, AS PUBLISHED (DIRECT HEAD), WITH PAGE, AND WITH PAGE PLUS THE SEIR BRANCH (THE SEIR-GNN, OR SEIR-LSTM FOR THE LSTM). SQUARED ERROR, NO SEASONAL FEATURES. BOTTOM: BASELINES;  $B$  IS THE CONFIGURATION VALIDATION SELECTS. SELECTION USES VALIDATION ONLY.

| encoder                   | direct            | PAGE          | PAGE+SEIR     |
|---------------------------|-------------------|---------------|---------------|
| AAGCN                     | 16.71 / 35.15     | 16.57 / 33.53 | 17.43 / 35.58 |
| ASTGCN                    | 16.83 / 35.61     | 16.76 / 34.76 | 17.51 / 35.71 |
| LSTM                      | 16.91 / 35.10     | 16.94 / 35.85 | 17.69 / 35.52 |
| STGAT                     | 22.93 / 62.00     | 16.98 / 36.46 | 17.63 / 35.84 |
| A3TGCN                    | 28.63 / 60.40     | 17.68 / 37.16 | 17.68 / 35.50 |
| DCRNN                     | 34.64 / 73.52     | 18.35 / 39.02 | 17.67 / 35.52 |
| baseline                  | validation / test |               |               |
| Persistence               | 17.86 / 36.02     |               |               |
| No message passing (res.) | 16.90 / 34.57     |               |               |
| GCN (res.)                | 16.84 / 34.50     |               |               |
| STID-style MLP [18]       | 16.35 / 36.33     |               |               |
| $k$ -NN analogues         | 16.52 / 34.34     |               |               |
| NB-GLM                    | 16.97 / 42.85     |               |               |
| Boosted trees             | 17.94 / 38.96     |               |               |
| Boosted trees + climate   | 17.40 / 36.49     |               |               |
| $B$ : AAGCN, NB, season   | 15.51 / 37.54     |               |               |

TABLE II

ABLATION: FROM THE PUBLISHED DIRECT HEAD TO PAGE. EACH ENCODER BEHIND FOUR HEADS THAT ADD, IN TURN, THE PERSISTENCE ANCHOR (RESIDUAL), THE LEARNED GATE (PAGE) AND THE SEIR BRANCH (MEAN VALIDATION RMSE; CONFIGURATION OF TABLE I). THE LAST TWO COLUMNS PAIR PAGE+SEIR WITH PAGE; NEGATIVE FAVOURS THE SEIR BRANCH, AND THE COUNT IS RUNS WHERE IT WINS.

| encoder     | direct resid. | PAGE  | SEIR branch |       |             |             |
|-------------|---------------|-------|-------------|-------|-------------|-------------|
|             |               |       | PAGE        | +SEIR | val         | test        |
| AAGCN       | 16.71         | 16.72 | 16.57       | 17.43 | +0.86 (1/9) | +2.05 (0/9) |
| ASTGCN      | 16.83         | 16.85 | 16.76       | 17.51 | +0.75 (0/9) | +0.95 (3/9) |
| LSTM        | 16.91         | 16.96 | 16.94       | 17.69 | +0.75 (0/9) | -0.33 (6/9) |
| STGAT       | 22.93         | 17.85 | 16.98       | 17.63 | +0.66 (1/9) | -0.62 (8/9) |
| A3TGCN      | 28.63         | 17.47 | 17.68       | 17.68 | -0.00 (6/9) | -1.66 (9/9) |
| DCRNN       | 34.64         | 17.52 | 18.35       | 17.67 | -0.68 (3/9) | -3.50 (9/9) |
| persistence | 17.86         |       |             |       |             |             |

seed) units and tested with an exact sign-flip permutation test, Benjamini–Hochberg adjusted. The prospective test trains once on all development weeks, freezes every setting, and scores 85 forecast starts from 2024-W11, with no target week shared between partitions; it uses a block bootstrap over forecast starts (block 8, 10,000 replicates) with Holm correction over two pre-registered comparisons. RMSE in weekly cases, pooled over windows, districts and horizons, is the primary metric; MAE, SMAPE and MAPE are secondary.

## VI. RESULTS

All numbers are RMSE in weekly cases. Development results are means over three origins  $\times$  three seeds; validation decides and test is reported beside it.

### A. Comparative Analysis

Table I compares every encoder and baseline on identical windows. **As published, three of five ST-GNNs fail.** Behind the direct head, AAGCN, ASTGCN and the LSTM reach 16.71–16.91 on validation and all beat persistence (17.86), but STGAT, A3TGCN and DCRNN reach 22.93, 28.63 and 34.64, worse than repeating last week. They underfit rather than lack information: a linear model fitted to their training residuals recovers 48% and 60% of STGAT’s and A3TGCN’s validation residual variance, against 3–7% for the working encoders.

**PAGE makes every encoder usable.** With PAGE, all six encoders score 16.57–18.35 on validation. Paired against the published head, it repairs STGAT, A3TGCN and DCRNN by 5.95, 10.95 and 16.29 on validation and by 25.54, 23.24 and 34.50 on test, winning 9/9 runs each ( $p_{\text{adj}} = 0.004$ ). On the working encoders it changes validation by at most 0.14, and it lowers AAGCN’s test RMSE on 9/9 runs (−1.62). **With a strong encoder it beats persistence:** AAGCN with PAGE is better than persistence on all nine runs on validation (−1.29) and on test (−2.48), and ASTGCN on 9/9 and 8/9 (−1.10, −1.26). The repaired encoders reach persistence but do not clearly pass it. AAGCN with PAGE has the lowest test RMSE of any configuration (33.53), but validation selects  $B$  (AAGCN, direct head, NB loss, seasonal features), which is worse than persistence on test (37.54), so we report AAGCN with PAGE as observed rather than selected.

The graph adds little: a GCN beats the same network without message passing by only 0.06 (3/9 runs), and  $k$ -NN analogues with no network at all reach 16.52 / 34.34.

### B. Ablation Study

Table II builds PAGE from the published head one component at a time. **The anchor removes the failure:** a residual head with no gate already brings STGAT, A3TGCN and DCRNN to 17.47–17.85. **The gate improves on it for STGAT** (17.85 to 16.98), AAGCN and ASTGCN, costs A3TGCN and DCRNN 0.21 and 0.83, and on the prospective weeks beats the ungated head (−0.31, 95% CI [−0.77, −0.08]).

**The SEIR branch adds nothing.** A pre-registered rule credited physics only if the SEIR branch beat PAGE on at least 7/9 runs with  $p_{\text{adj}} < 0.05$  on each failing encoder. It does so on none: it is *worse* on STGAT (+0.66, 1/9), ties on A3TGCN (−0.00, 6/9), helps DCRNN on only 3/9 runs, and costs the working encoders 0.75–0.86 (0–1/9). Routing the whole forecast through the simulator is worse still: the pure SEIR decoder scores 24.01–25.75, worse than persistence on every encoder. Inverting the simulator explains why. With transmission set to zero, the exposed state seeded from recent cases already emits more cases than observed for 14–16% of training targets, which no network output can reach, and the required force of infection is only  $r^2 = 0.22$ –0.26 predictable from case history.

**Loss.** Among more than twenty further levers, each paired against its own control, the NB likelihood gives the largest validation gain (−0.76, 9/9,  $p_{\text{adj}} = 0.014$ ), larger than any architectural change, but it is 1.61 worse on test, so PAGE

TABLE III

PROSPECTIVE EVALUATION ON WEEKS NO MODEL OR SETTING HAD SEEN: 85 FORECAST STARTS, 2024-W11 TO 2026-W32. SETTINGS FROZEN ON DEVELOPMENT DATA BEFORE THE NEW WEEKS WERE PARSED. NEURAL ARMS: MEAN OVER 3 SEEDS, SEED RANGE OF RMSE IN BRACKETS. BOLD: BEST PER METRIC. PRE-REGISTERED TESTS (BLOCK BOOTSTRAP OVER FORECAST STARTS, HOLM OVER TWO): P1, SEIR BRANCH VS PAGE:  $-0.03 [-0.41, +0.12]$ ,  $p_{\text{HOLM}} = 0.67$ ; P2, PAGE+SEIR VS PERSISTENCE:  $+0.65 [-0.99, +1.69]$ ,  $p_{\text{HOLM}} = 0.62$ .

| arm                     | RMSE                 | MAE          | SMAPE       | MAPE        | RMSE $h=1/2/3$ |
|-------------------------|----------------------|--------------|-------------|-------------|----------------|
| Persistence             | <b>34.75</b>         | 11.78        | 50.3        | 53.3        | 15.4/35.7/46.0 |
| PAGE (adaptive graph)   | 35.43 [35.24, 35.60] | 11.35        | <b>45.7</b> | 43.9        | 16.0/35.7/47.3 |
| PAGE + SEIR branch      | 35.40 [35.11, 35.56] | <b>11.29</b> | 46.1        | 43.5        | 15.8/35.7/47.3 |
| Residual head (no gate) | 35.74 [35.37, 36.19] | 11.61        | 46.6        | <b>43.0</b> | 16.4/36.0/47.6 |
| AR(3) ridge             | 36.35                | 11.35        | 46.7        | 44.5        | 17.3/35.9/48.8 |
| Seasonal naive          | 62.43                | 27.53        | 74.6        | 113.4       | 50.7/62.5/72.2 |

TABLE IV

COMPUTATIONAL COST. PARAMETERS OF THE FULL MODEL BEHIND THE DIRECT HEAD, AND THE PARAMETERS EACH HEAD ADDS. SECONDS PER EPOCH AND SECONDS PER TRAINING RUN (WITH EARLY STOPPING) ARE MEANS OVER THE 9 RUNS OF THE KAGGLE RUN, EACH RUN SINGLE-THREADED ON A 4-CORE KAGGLE CPU. PAGE ADDS ONE PARAMETER AND NO MEASURABLE TIME; THE SEIR BRANCH ADDS FIVE PARAMETERS AND ABOUT  $2\times$  THE TIME PER EPOCH ON THE LIGHT ENCODERS.

| encoder | parameters |      |       | s / epoch |      |       | s / run |
|---------|------------|------|-------|-----------|------|-------|---------|
|         | direct     | PAGE | +SEIR | direct    | PAGE | +SEIR | PAGE    |
| AAGCN   | 2,873      | +1   | +5    | 0.19      | 0.20 | 0.41  | 19      |
| ASTGCN  | 50,769     | +1   | +5    | 0.92      | 0.97 | 1.16  | 62      |
| LSTM    | 50,627     | +1   | +5    | 0.46      | 0.49 | 0.68  | 35      |
| STGAT   | 160,894    | +1   | +5    | 0.23      | 0.26 | 0.42  | 14      |
| A3TGCN  | 8,742      | +1   | +5    | 0.23      | 0.23 | 0.36  | 14      |
| DCRNN   | 827,843    | +1   | +5    | 4.08      | 3.85 | 3.56  | 220     |

keeps squared error. No architecture change improves validation by more than 0.15, and RevIN (+1.47), an STID-style MLP (+0.84) and TimeGAN augmentation (+0.60) are worse on 0/9 runs.

### C. Prospective Evaluation, 2024–2026

The decisive test uses weeks on which no setting was chosen. Six arms were frozen on development data before the new weeks were parsed, and each was scored once on 85 forecast starts (Table III). The neural arms share the development finalist encoder, a Graph WaveNet-style network with a learned adjacency [8].

**No learned model beats persistence on RMSE:** persistence scores 34.75 and PAGE 35.43, a difference that is not significant ( $p_{\text{Holm}} = 0.62$  for the SEIR variant). **The SEIR branch ties PAGE** ( $-0.03$ ,  $[-0.41, +0.12]$ ,  $p_{\text{Holm}} = 0.67$ ). On the secondary metrics PAGE is better than persistence (MAE 11.35 against 11.78, SMAPE 45.7 against 50.3), and every arm beats seasonal naive by 26–28 RMSE. The 2026 outbreak dominates RMSE: persistence scores 14.72, 20.05 and 160.51 in 2024, 2025 and 2026, and PAGE 14.32, 21.43 and 162.75. PAGE is better in the quiet year and worse in the outbreak, whose next large move is not in the inputs. The SEIR arm’s development advantage over persistence (27.34 against 28.54) does not carry over. Because the two variants are equivalent,

we present the simpler PAGE; this choice was made after the test, and the pre-registered finalist was the SEIR variant.

### D. Hyperparameter Tuning and Computational Cost

Every encoder and head in the main study uses the same fixed settings (Section IV-F), so no model receives more tuning than another. Before the prospective test, each learned arm was tuned on the same grid (learning rate  $\{10^{-3}, 3 \times 10^{-3}\}$   $\times$  hidden size  $\{32, 64\}$ , 108 runs per arm) on validation only; all arms chose  $3 \times 10^{-3}$  and 64, and the grid moved validation RMSE by at most 0.57. Table IV gives the cost. PAGE adds one parameter and no measurable time, while the SEIR branch roughly doubles the time per epoch on the light encoders because it steps the simulator daily. The smallest encoder, AAGCN (2,873 parameters, 19 s per run), is also the most accurate.

## VII. DISCUSSION

**Why PAGE works.** Weekly dengue counts are close to a random walk, so the most informative feature is the last value. A direct head must carry that value through the encoder, and three published architectures, attention-based, recurrent and diffusion-based, lose it. PAGE hands the level to the output, so the encoder only learns the departure, and the gate stops a weak departure from pulling the forecast away from persistence. This is why one parameter repairs three different designs.

**What it cannot do.** The anchor removes failure but cannot create information. The working encoders, the SEIR-LSTM and  $k$ -NN, which has no network at all, leave residuals that correlate 0.91–0.97 with one another: the models make the same errors, and those errors are lags on large moves. When the next move is not in the inputs, as in the 2026 outbreak, every model lags it, and on later weeks no learned model beats persistence on RMSE.

**Physics.** The SEIR simulator cannot be the forecaster, because 14–16% of targets lie below its zero-transmission floor and its force of infection is poorly predictable. Inside the gated head it looked useful, but the ablation and the prospective test both show that the anchor and gate do the work. For this target, the useful prior is persistence, not transmission dynamics. We therefore recommend that dengue forecasting studies report

persistence, a matched no-physics control and, where possible, a prospective test.

**Limitations.** One country, one disease and district-week resolution; three origins measure run-to-run stability more than the series; the prospective test is amended, covers one 2.4-year period dominated by one outbreak, and uses one encoder family; and the graph is the benchmark’s hand-built adjacency rather than mobility.

## VIII. CONCLUSION

We proposed PAGE, a persistence-anchored gated head that plugs into any spatio-temporal encoder at the cost of one parameter, together with a corrected, leakage-free Sri Lankan dengue dataset extended to 2026 with newly parsed reports. PAGE turns three published ST-GNNs that are worse than persistence into usable forecasters, and with a strong encoder it beats persistence on every paired development run. The ablation shows that the gain comes from the anchor and the gate, not from the SEIR simulator. On 2024–2026 weeks no learned model beats persistence on RMSE, although PAGE is better on MAE and SMAPE. Future gains must come from inputs that anticipate outbreaks, not from the architecture.

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